

Design via root locus

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Mechatronics 3DX4

Slides 9: Design Via Root Locus

Dr. O. Onaizah

Department of Computing and Software
McMaster University

Material based on *Control Systems Engineering* by N. Nise (9.1-9.4).

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Introduction

- ▶ Root locus graphically displays both transient response and stability information.
- ▶ The root locus allows us to choose the needed gain to get the desired transient response.
- ▶ **Problem:** when we choose different gains, we can only achieve responses that correspond to roots that are actually on the root locus.
- ▶ By adding a controller in series with the original system, we can add new poles to the system, located at the desired spot.

只调节constant gain K 只能把当前系统的 G 的极点沿着root locus调整，所以我们需要打破这个限制，必须串联一个控制器来增加新的pole和zero的位置。

Improving Transient Response

- ▶ If point B in Fig. 9.1 represented our desired transient response, represented by desired percent overshoot and settling time, we are out of luck.
- ▶ Using gain response, we can achieve the desired percent overshoot, but *not* the settling time at point A.

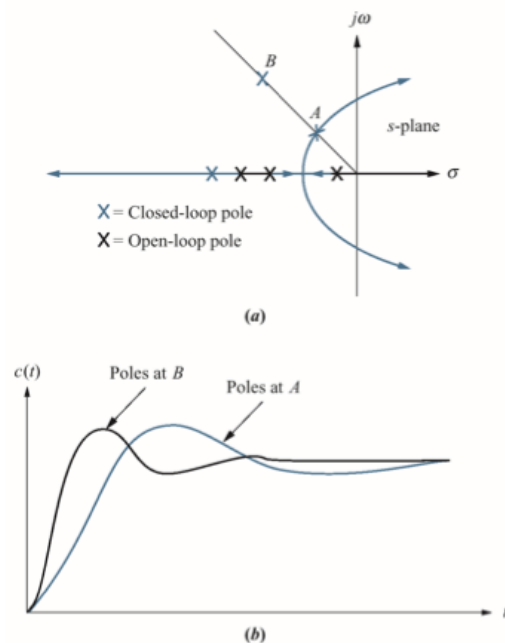


Figure 9.1

Improving Transient Response II

- ▶ Replacing original system would be expensive and may be difficult to find system with correct response and still satisfy other needed properties such as speed, power, durability etc.
- ▶ Instead we augment, or **compensate**, system by adding additional poles and zeros to get desired behavior.
- ▶ **Disadvantage:** Method increases order of system, which can also affect the response.
- ▶ Our design process will determine the location of the

- ▶ Our design process will determine the location of the second-order closed-loop poles but not the higher-order ones.
- ▶ Must always evaluate response via simulation at end to ensure that actual higher-order system behaves as desired.

我们这里说的用compensator去补偿，得到额外的pole和zero，但是这会增加系统的order。

Improving Steady State Error (SSE)

- ▶ Can also add compensators to improve steady state error.
- ▶ Typically, increasing gain produces smaller steady state error, but larger overshoot.
- ▶ When we optimize gain for transient response, we can make steady state error worse.
- ▶ We can improve steady state error separate from gain by adding a compensator with an integrator ($\frac{1}{s}$) in feed forward path.

通常调大增益K能减小误差，但会导致OS%变大（瞬态变差）。通过在正向通道加一个积分器（ $1/s$ ），可以独立改善稳态误差而不影响增益。

Compensator Terminology

- ▶ **Proportional control systems** are compensators that feed the error signal directly to the plant.
- ▶ **Integral control systems** feed the integral of the error to the plant.
- ▶ **Derivative control systems** feed the derivative of the error to the plant.

比例控制（P）直接反馈误差；积分控制（I）反馈误差的积分；微分控制（D）反馈误差的微分。

Configurations

- ▶ Two main configurations are **cascade compensators** and **feedback compensators**.
- ▶ Both methods can be used to change the open-loop poles and zeros location, and thus the root locus.

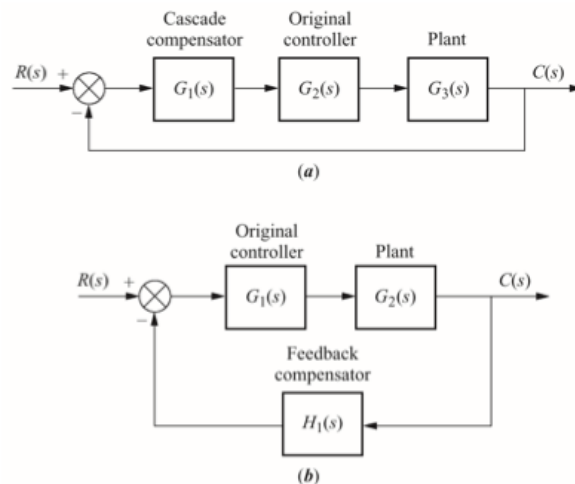


Figure 9.2.

补偿器两种放置方式，forward或者feedback放置。

Improving SSE via Cascade Compensation

- ▶ Two approaches: **ideal integral compensation** and **lag compensation**.
- ▶ First method will produce zero steady state error, but requires active circuit such as an op amp circuit (if implemented in analog).
- ▶ Second method will only reduce the error, but can be realized using passive circuits (if implemented in analog).

Note: If the compensator is done in software there is only an increase in complexity of the software, not the system hardware!

改善稳态误差有两种方法：理想积分补偿（主动电路，能让误差归零）和滞后补偿（无源电路，只能减小误差）。

Ideal Integral Compensation

- ▶ Method involves adding an integrator ($\frac{1}{s}$) in feed forward path.
- ▶ Increases system type, causing error to go to zero.
- ▶ As implementation of method consists of both feeding the error and its integral to the plant, we use term **proportional-plus-integral (PI) controller**.

直接在主路上加一个位于原点的极点 $1/s$ ，把系统类型（Type）提升一级，误差变为 0。这就是PI控制。

Ideal Integral Compensation: II

- ▶ Simply adding a pole at the origin would change the transient response of the original system.
- ▶ We see in example below, the desired pole location at A is now no longer on the root locus.

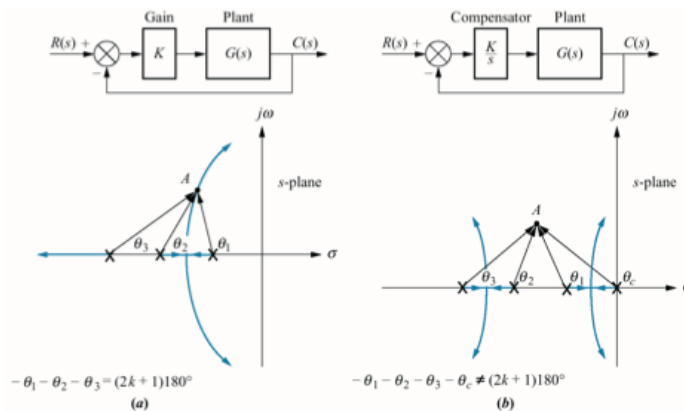
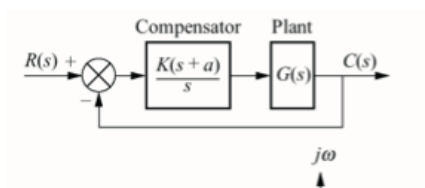


Figure 9.3.

凭空加一个极点，会破坏原有的角度条件，导致根轨迹整体向右偏移（系统变慢甚至不稳定），原来想要的极点 A 不在轨迹上了。

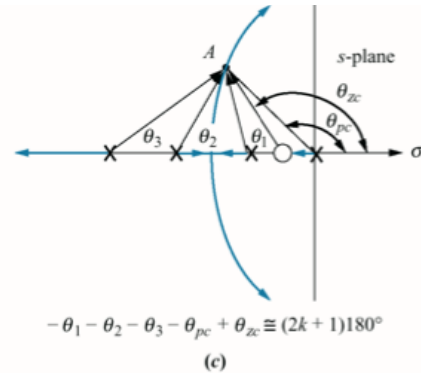
Ideal Integral Compensation: Add Zero too!

- ▶ Fix this by adding a zero near the origin at $s = -a$.



- ▶ The angular contribution of the two effectively cancel.
- ▶ Require gain (K) is about the same as the ratio of the pole and zero magnitudes is about unity.

Figure 9.3



解决办法! 既然极点破坏了角度, 我们就紧挨着原点 (比如 $S=-0.1$) 再加一个零点。极点和零点挨得很近, 它们对远处极点 A 的角度贡献就相互抵消了, 根轨迹又恢复了原样。注意这种情况下compensator可以写成 $K + \frac{K*a}{s}$ 。

Ideal Integral Compensation: Implementation

- ▶ Can implement controller as

$$G_c(s) = K_1 + \frac{K_2}{s} = \frac{K_1 s}{s} + \frac{K_2}{s} = \frac{K_1(s + \frac{K_2}{K_1})}{s}$$

- ▶ Example of a proportional-plus-integral controller.

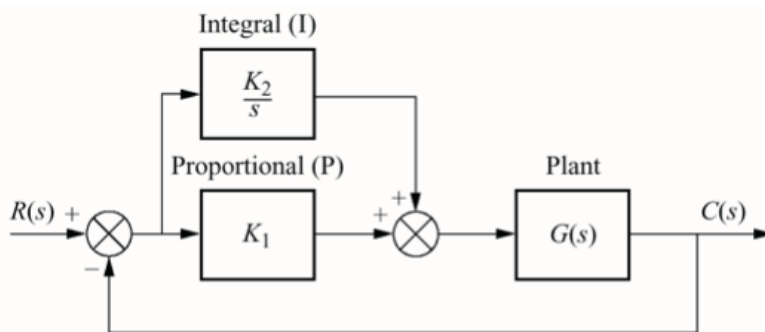


Figure 9.8.

Effect of Ideal Integral Compensation

- In example below, can we add the compensator shown to reduce SSE to zero but have small effect on the transient response?

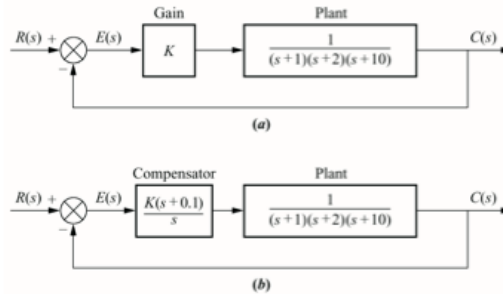


Figure 9.4.

Effect of Ideal Integral Compensation II

- Figure shows uncompensated system and desired dominant poles.
- Specification was for damping ratio of 0.174, thus angle $100.02^\circ = 180^\circ - \cos^{-1}(0.174)$.
- System has

$$\begin{aligned} K_p &= \lim_{s \rightarrow 0} KG(s) \\ &= \frac{164.6 \times 1}{1 \times 2 \times 10} = 8.23 \end{aligned}$$

which gives $e_{ss} = \frac{1}{1+K_p} = 0.108$.

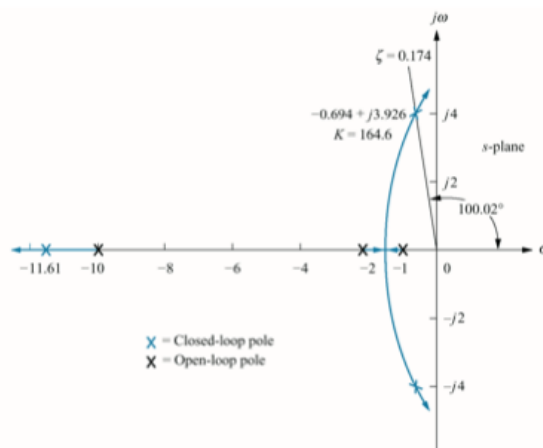


Figure 9.5

这里用了公式 $s = -\sigma + i\omega_d$ 来解析描述pole，然后斜边长度是 ζ ，求解出角度（负实轴和 $j\omega$ 的夹角）

Effect of Ideal Integral Compensation III

- ▶ Figure shows system with added compensator.
- ▶ System responds to step input with zero error. Has gain and original three poles very close to previous values.
- ▶ Fourth closed loop pole at $s = -0.0902$ so essentially cancels with zero at $s = -0.1$.

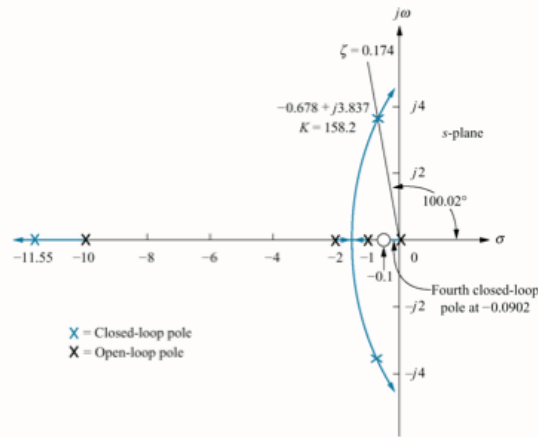


Figure 9.6

Effect of Compensation: Output Simulation

- ▶ Figure shows step response of original and compensated system.
- ▶ Original system takes 6 seconds to reach $\pm 2\%$ of final value while compensated system takes 18 seconds.
- ▶ Compensated system reaches $\pm 2\%$ of original final value in about same time. Extra time is to reach new final value.

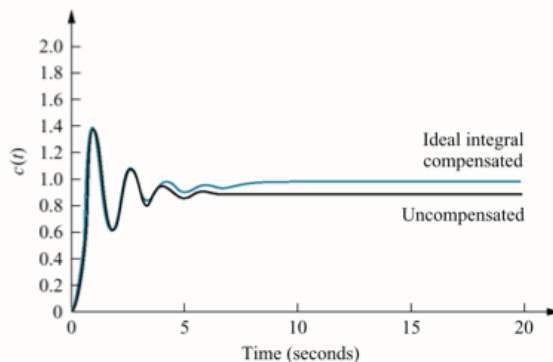


Figure 9.7.

加了补偿之后, e_{ss} 完全被消除, 但是settling time变长了。

Lag Compensation

- ▶ Require active circuit to put pole at origin.
- ▶ Passive circuits can only put a pole near origin, but can still improve steady state error.

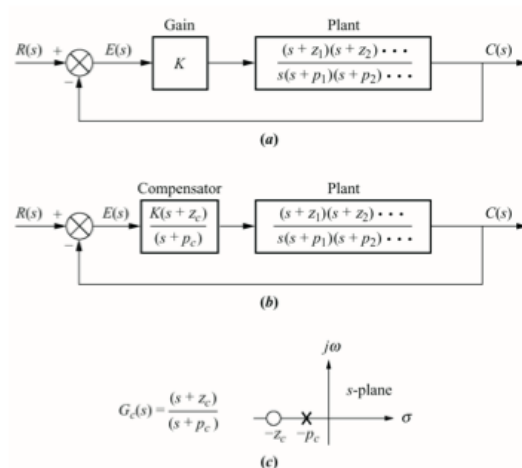
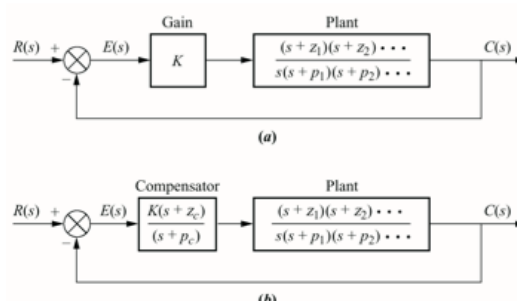


Figure 9.9

加 $1/s$ 这种行为本质上是加几个active components, 比如说opamp, 需要额外供电。我们现在找的是替代方案, 用passive filter达到类似的效果, 这就是lag compensation。

Lag Compensation II

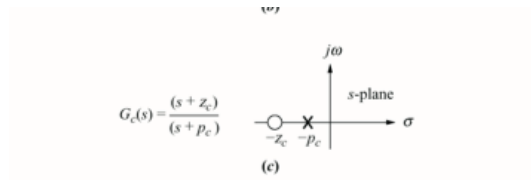
- ▶ For type one system shown, $e_{ss} = \frac{1}{K_v}$.
- ▶ For original system we have $K_{vO} = \frac{K z_1 z_2 \dots}{p_1 p_2 \dots}$.
- ▶ For compensated system, we have $K_{vN} = \frac{(K z_1 z_2 \dots)(z_c)}{(p_1 p_2 \dots)(p_c)} =$



$$K_{vO} \times \frac{z_c}{p_c}.$$

- ▶ Increasing K_v will decrease SSE.

Figure 9.9



Lag Compensation III

- ▶ New steady state error is $e_{ssO} \times \frac{p_c}{z_c}$.
- ▶ Need to maximize $\frac{z_c}{p_c}$ ratio to reduce SSE, but need the zero and pole to be close together so their effect cancels.
- ▶ Means z_c and p_c must both be small, thus close to the origin.

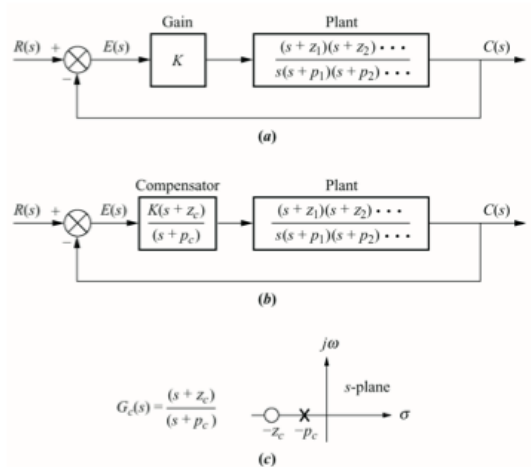


Figure 9.9

这里是推导lag compensation的ess, $e_{ss} = e_{ssO} * \frac{p_c}{z_c}$, 我们要最小化ess。

Improving Transient Response via Compensation

- ▶ Goal: design system with desired percent overshoot, but better settling time than uncompensated system.
- ▶ Two approaches: **ideal derivative compensation** and **lead**

- ▶ Two approaches: **ideal derivative compensation** and **lead compensation**.

- ▶ Involves adding a zero (differentiator) to the forward path.
- ▶ Differentiation is noisy. Can add a large unwanted signal.

这里我们关注怎么让系统反应更快，即减少settling time，但是保持OS%不变。两种方法：ideal derivative compensation & lead compensation。

Ideal Derivative Compensation

- ▶ Adding only a zero to forward path requires active circuit.
- ▶ As implementation of method consists of both feeding the error and its derivative to the plant, we use term **proportional-plus-derivative (PD) controller**.
- ▶ Goal is to reshape root locus so that desired closed loop pole is on root locus.

Ideal derivative就是直接在forward path上加一个zero，但是这需要active component。这就是PD控制。

Ideal Derivative Compensation II

- ▶ To add a zero to forward path, we add compensator of form:

$$G_c(s) = s + z_c$$

- ▶ Choosing the right location for the zero can quicken the response of the original system.
- ▶ For step input and position control, system responds to sudden large change in input.
- ▶ This means that the derivative of this quick change will produce an even larger signal, driving the system even faster forward.

PD 控制的物理意义：它不仅看当前的误差，还看误差的变化率（导数）。当误差剧烈变化时，它会给系统一脚“猛油门”，让系统动作更快。

Ideal Derivative Compensation Design

- ▶ We first identify location of closed-loop poles with desired overshoot and settling time.
- ▶ The difference between 180° and $\angle KG(s)H(s)$ to this pole location is the needed angular contribution of the compensator zero.
- ▶ For system below, design PD controller to yield 16% overshoot, and a threefold reduction in settling time.

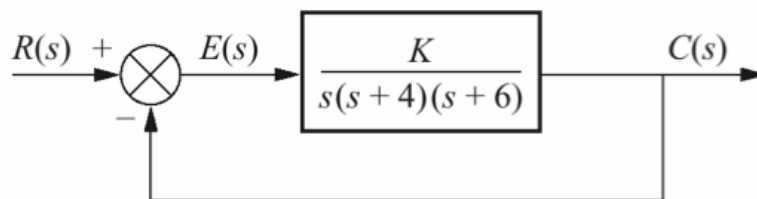


Figure 9.17.

Ideal Derivative Compensation Design II

- ▶ 16% overshoot corresponds to $\zeta = 0.504$, thus angle $120.26^\circ = 180^\circ - \cos^{-1}(0.504)$.
- ▶ Searching along this line of damping ratio discovers closed loop poles at $-1.205 \pm j2.064$ with gain $K = 43.35$.
- ▶ Settling time of system is $T_s = \frac{4}{\zeta\omega_n} = \frac{4}{1.205} = 3.320$.

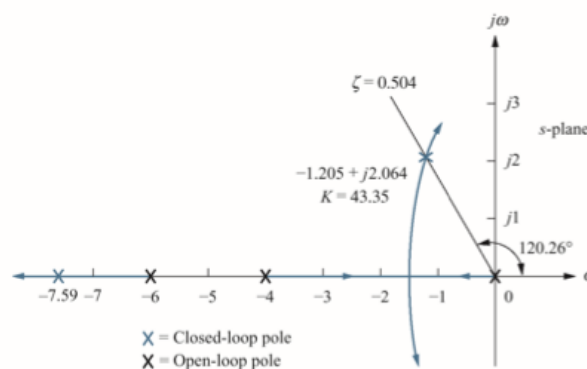


Figure 9.18.

Ideal Derivative Compensation Design III

- ▶ Gain of $K = 43.35$ gives us a third closed loop pole at $s = -7.59$ which is more than six times farther from the $j\omega$ axis than dominant poles.
- ▶ Our 2nd order approximation is thus valid for overshoot and settling time.

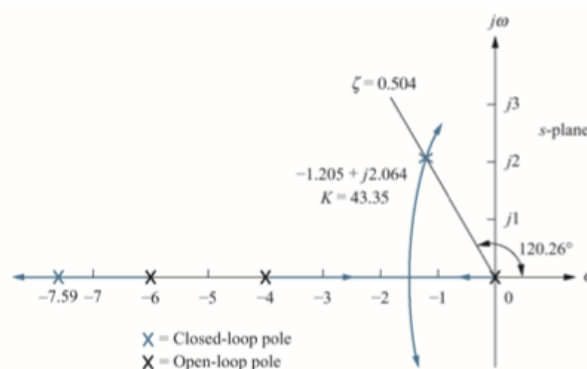


Figure 9.18.

Ideal Derivative Compensation Design IV

- Desired settling time is $\frac{3.320}{3} = 1.107$.
- Real part of desired pole is $\sigma = \frac{4}{T_s} = \frac{4}{1.107} = 3.613$.
- Imaginary part is $\omega_d = 3.613 \tan(180^\circ - 120.26^\circ) = 6.193$.

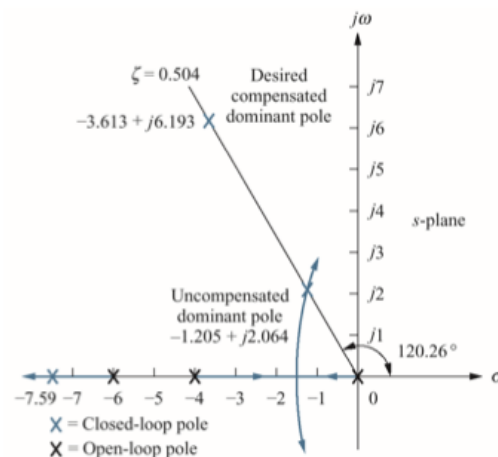
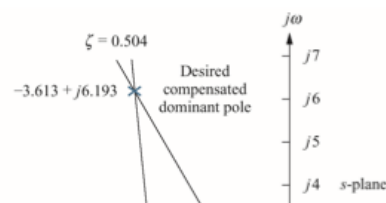


Figure 9.19.

Uncompensated dominant pole就是当没有加补偿器的时候，系统的pole如果想满足 ζ 的条件只能落在这，但是settling time比较大，我们想要减少 T_s 只能用补偿器给pole挪动到我们desired的位置。

Ideal Derivative Compensation Design V

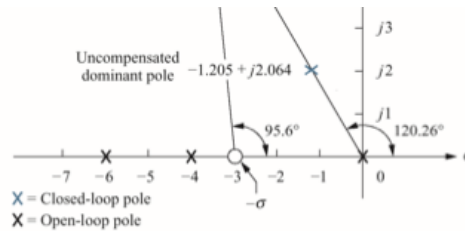
- For a point to be on locus, must satisfy $\angle KG(s)H(s) = (2k + 1)180^\circ$.
- For $s = -3.613 + j6.193$, we get sum of angles to be -275.6° .
- Zero must contribute $\theta = 275.6^\circ - 180^\circ = 95.6^\circ$.
- Location of zero must satisfy $\tan(180^\circ - 95.6^\circ) = \frac{\text{opp}}{\text{adj}} =$



$$\frac{6.193}{3.613 - \sigma}$$

- We thus need a zero at $s = -3.006$.

Figure 9.20



但是我们必须得满足角度条件，我们desired的点角度不符合，所以我们必须得添加一个zero来满足角度条件。

Ideal Derivative Compensation Design VI

- To get desired dominant poles, we need a gain of $K = 47.54$.
- At this gain, we get a third pole at $s = -2.755$.
- Second order approximation may not be valid as not close enough to get good pole-zero cancellation.

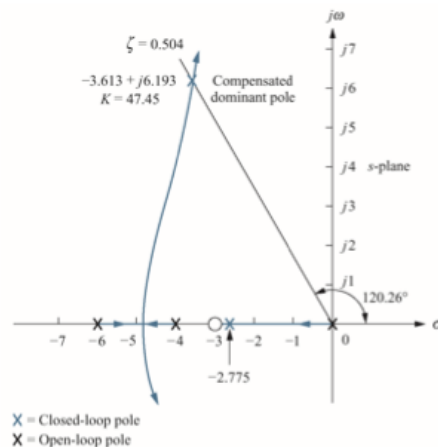


Figure 9.21.

这里第三个pole离得不够远，二阶近似可能失效。

Ideal Derivative Compensation Design VII

	Uncompensated	Simulation	Compensated	Simulation
Plant and compensator	K		$K(s + 3.006)$	

	$s(s+4)(s+6)$		$s(s+4)(s+6)$	
Dominant poles	$-1.205 \pm j2.064$		$-3.613 \pm j6.193$	
K	43.35		47.45	
ζ	0.504		0.504	
ω_n	2.39		7.17	
%OS	16	14.8	16	11.8
T_s	3.320	3.6	1.107	1.2
T_p	1.522	1.7	0.507	0.5
K_v	1.806		5.94	
$e(\infty)$	0.554		0.168	
Third pole	-7.591		-2.775	
Zero	None		-3.006	
Comments	Second-order approx. OK		Pole-zero not canceling	

Table 9.3.

Ideal Derivative Compensation Design VIII

- Figure shows step response of uncompensated and compensated system.

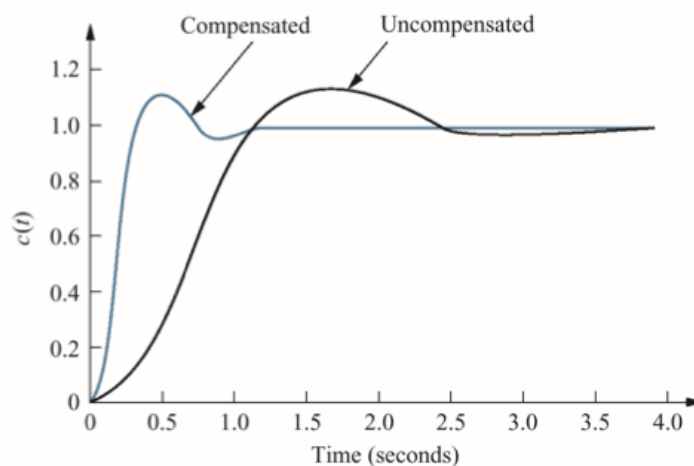


Figure 9.22.

PD Controller

- We implement an ideal derivative controller as a PD controller shown below.

- The transfer function of the compensated system is

- ▶ It has transfer function

$$G_c(s) = K_2s + K_1 = K_2\left(s + \frac{K_1}{K_2}\right)$$

- ▶ Choose K_2 to contribute to the required gain, and choose $\frac{K_1}{K_2}$ to position the zero at the correct place.

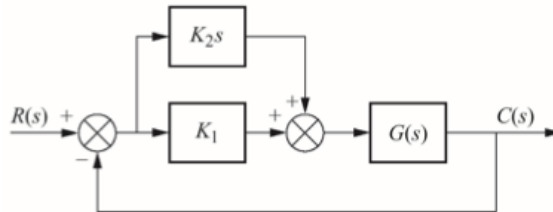


Figure 9.23.

Lead Compensation

- ▶ Advantages of a passive lead compensator
 - ▶ doesn't require additional power supply
 - ▶ reduced noise due to differentiation
- ▶ Approximates ideal derivative compensator by adding a zero and a pole.
- ▶ As long as pole farther to the left than zero, net angular contribution is still positive.

类似于我们之前在PI控制那里讲的，一种折中方案，不适用active component，用lead compensation。

Lead Compensation II

- ▶ As for a PD controller, we first determine the location of the closed loop dominant poles with desired properties.
- ▶ The angular contribution of zero, z_c , is θ_2 and contribution of pole, p_c , is θ_1 .
- ▶ Net contribution of pole-zero pair is thus $\theta_c = \theta_2 - \theta_1$.
- ▶ Choose θ_1 and θ_2 such that $\theta_2 - \theta_1 - \theta_3 - \theta_4 + \theta_5 = (2k + 1)180^\circ$

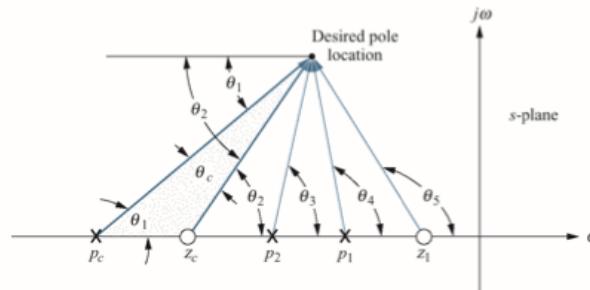


Figure 9.24.

和理想PD控制器不一样，用 passive 会贡献一个 pole 和 zero，但是 pole 更远靠左，这导致了 zero 贡献的角度大于 pole，也可以达到 root locus 的角度要求，也可以达到减少 T_s 的目的。

Lead Compensation III

- ▶ Have infinite possible pole-zero combinations that would work.
- ▶ Compensators differ in such things as static error constants, required gain, quality of 2nd order approximation, and the shape of actual transient response.

Figure 9.25
Three of the infinite possible lead compensator solutions

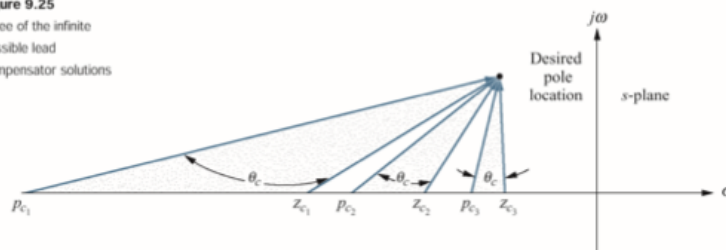


Figure 9.25.

和理想PD不一样，lead compensation有无数个解。

1. Lead Compensation (超前补偿) —— “让相位跑在前面”

“Lead” 的意思是**超前**。

- **物理表现**：如果你给这个补偿器输入一个正弦波，你会发现输出波形的波峰比输入波形的波峰**更早出现**。
- **数学本质**：它在特定频率范围内提供了**正的相位角 ($\Phi > 0$)**。
- **为什么**？因为它的零点 z_c 离原点近，极点 p_c 离原点远。在频域里，零点产生“相位上升”，极点产生“相位下降”。由于零点先起作用，所以总相位是往上拉的。
- **形象理解**：它就像是给系统喝了咖啡，让它提前进入状态，反应变快。

2. Lag Compensation (滞后补偿) —— “让相位落在后面”

“Lag” 的意思是**滞后**。

- **物理表现**：输入一个正弦波，输出波形的波峰会比输入波形的波峰**晚一点出现**。
- **数学本质**：它在特定频率范围内产生了**负的相位角 ($\Phi < 0$)**。
- **为什么**？它的极点 p_c 离原点更近。极点先起作用，把相位往下拽，所以相位落后了。
- **形象理解**：它更像是一个“减震器”或低通滤波器。它故意让高频反应变慢（滞后），从而换取低频段极高的增益，用来把稳态误差压下去。

Improving SSE and Transient Response

- ▶ To improve both transient response and steady state error, we can:
 - ▶ First design an active PD controller, and then an active PI controller. Called a **proportional-plus-integral-plus-derivative (PID) controller**.
 - ▶ First design a passive lead compensator, then design a passive lag compensator. Called a **lead-lag compensator**.

这就是PID。我们一般先做PD，再做PI。

PID Controller

- ▶ Transfer function of controller is

$$G_c(s) = K_1 + \frac{K_2}{s} + K_3s = \frac{K_3(s^2 + \frac{K_1}{K_3}s + \frac{K_2}{K_3})}{s}$$

- ▶ System has two zeros plus a pole at origin.

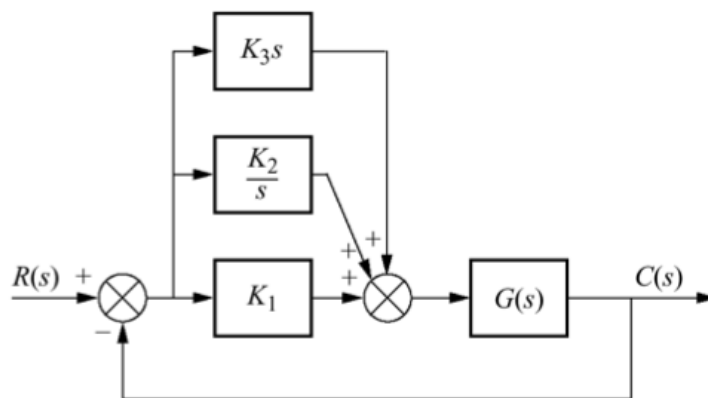


Figure 9.30.

PID Controller Design

1. Evaluate performance of original system.
2. Design PD controller for transient response improvement.
3. Simulate system to be sure requirements met.
4. Redesign if simulation shows they are not.

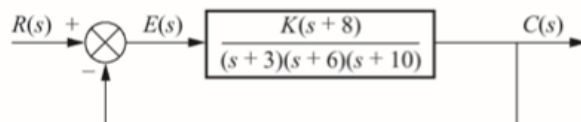
5. Design PI controller for steady state error requirement.
6. Determine gains K_1 , K_2 , K_3 .
7. Simulate system to be sure all conditions met.
8. Redesign if not met.

做题步骤:

1. 根据给的OS%算出 ζ ，画出射线，算出射线角度，判断input类型和ess的要求，以及对系统特性（比如说Ts或者Tp）的要求。
2. 找到当前系统和射线交点，算出需要的特性（比如说Tp），然后算出desired value，找到需要dominant pole移动到的新位置。
3. 先处理PD部分，把pole的新位置的数据带入G，算出角度和|G|，以及constant gain K，看看和 $2(k+1)*180$ 还差多少，然后加一个zero来满足这个条件。然后根据 $\arctan(\phi)$ ， ϕ 就是我们刚刚算的zero的角度，算出zero的x坐标，此时增益器是 $s + s_z$ 。
4. 然后处理PI，一般我们就是选一个靠近0点的位置，比如说a，然后增益器乘以 $\frac{s+a}{s}$ 。
5. 确定了增益器 G_c 之后，新的等效 G_e 就是 $G*G_c$ 。然后我们让 $1 + G_e*H*K = 0$ ，算出最终的constant gain K。然后我们化简增益G，判断系统参数。

PID Controller Design Example

1. Design PID controller so system has 20% overshoot, zero steady state error for step input, and peak time that is $\frac{2}{3}$ that of original system.

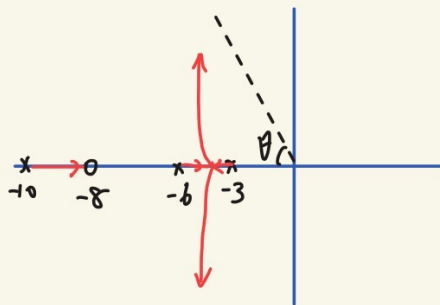


OS% = 20%, 0 ess for step input, $\frac{2}{3}$ peak time of original

Type 0 system

$$\zeta = \frac{-\ln(\%OS/100)}{\sqrt{\pi^2 + \ln^2(\%OS/100)}} = \frac{-\ln(0.2)}{\sqrt{\pi^2 + \ln^2 0.2}} = 0.456$$

$$\zeta = \cos \theta, \theta = 62.87^\circ$$



Dominant pole of original

$$-5.415 \pm 10.57j \text{ with } k = 121.5$$

Third pole at -8.169

$$\therefore 1 + k \cdot H \cdot G = 0 \therefore k = \frac{1}{|G|}$$

T_p of uncompensated system:

$$T_p = \frac{\pi}{\omega_d} = \frac{\pi}{10.57} = 0.297$$

Desired T_p is $\frac{2}{3}$ of original T_p

$$\omega_d = \frac{\pi}{\frac{2}{3} \times 0.297} = 15.87$$

$$\tan \theta = \frac{\omega_d}{\sigma} = \tan 62.87^\circ, \sigma = 8.13$$

We want dominant pole to be $-8.13 \pm 10.57j$

$$k = \frac{1}{|G|} = 5.34$$

$$G(s) = \frac{s+8}{(s+3)(s+6)(s+10)}, \text{ sub in } s_d$$

$$= -0.0035 + 0.0012j$$

$$\angle G(s_d) = \arctan\left(\frac{0.0012}{-0.0035}\right) = -198.37^\circ$$

$$\therefore \text{The zero angle contribution} = -198.37 + \theta_z = -180 \therefore \theta_z = 18.37^\circ$$

$$\tan \theta_z = \frac{\omega_d}{z_c - \sigma_d} = \frac{15.87}{z_c - 8.13} \therefore z_c = 55.92 \text{ a zero}$$

$$\therefore \text{Our PD controller is thus: } G_{PD} = s + 55.92$$

Type 0 system, so ess for unit step input & const. for 0 error, we need type 1, so add an integrator $\frac{1}{s}$, and we need cancel extra pole effect, so add a zero at -0.5,

$$G_{PI} = \frac{s+0.5}{s}$$

$$\therefore G_{PID}(s) = \frac{k(s+0.5)(s+55.92)}{s}, \quad k = 4.6$$

$$= \frac{4.6(s^2 + 56.42s + 27.96)}{s}$$

$$\therefore k_1 = 259.5, \quad k_2 = 128.6, \quad k_3 = 4.6$$


```

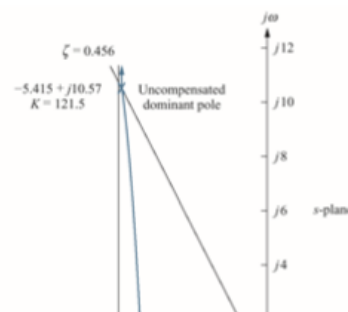
s = tf('s');
G=(s+8)/((s+3)*(s+6)*(s+10));
POS=20;
zeta=-log(POS/100)/sqrt(pi^2+(log(POS/100))^2);
disp(zeta);
theta=acos(zeta);
disp(rad2deg(theta));
% rlocus(G)
% sgrid(zeta,[])
s_wo=-5.42+10.6j;
T_po=pi/10.6;
T_p=2*T_po/3;
w_d=pi/T_p;
sigma_d=w_d/tan(theta);
disp(sigma_d);
s_w=-8.1456+15.71j;
system_angle=angle(evalfr(G,s_w));
disp(rad2deg(system_angle));
angle_PD=180-rad2deg(system_angle);
disp(angle_PD);
z=-8.15-15.71/tan(angle_PD);
disp(z);
G_c=(s+55)*(s+0.1)/s;
G_o=G_c*G;
k_final=1/abs(evalfr(G_o,s_w));
disp(k_final);

```

PID Controller Design Example II

- ▶ 20% overshoot gives us $\zeta = 0.456$ at a line at 117.13° .
- ▶ Gives dominant poles at $s = -5.415 \pm j10.57$, third pole at $s = -8.169$, and estimated peak time of 0.297 seconds.
- ▶ Desired pole has imaginary part

$$\omega_d = \frac{\pi}{T_p} = \frac{\pi}{(2/3)(0.297)} = 15.87.$$
- ▶ Real part is thus at $\sigma = \frac{\omega_d}{\tan(180^\circ - 117.13^\circ)} = 8.13$.
- ▶ Our desired dominant poles are thus at



$s = -8.13 \pm j15.87$, which requires gain $K = 5.34$.

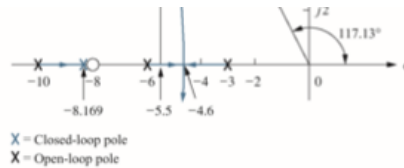


Figure 9.32

PID Controller Design Example III

- ▶ Sum of angles to $s = -8.13 + j15.87$ is currently -198.37° .
- ▶ Zero contribution is thus $\theta_z = 198.37^\circ - 180^\circ = 18.37^\circ$.
- ▶ This means we need a zero at $s = -55.92$.

- ▶ Our PD controller is thus $G_{PD} = (s + 55.92)$.

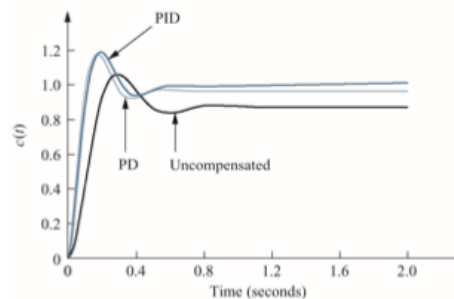


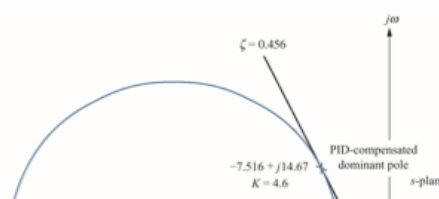
Figure 9.35

PID Controller Design Example IV

- ▶ We choose our PI controller to be $G_{PI} = \frac{(s+0.5)}{s}$
- ▶ This gives dominant poles at $s = -7.516 \pm j14.67$ at gain $K = 4.6$.
- ▶ Combining PI and PD to create PID controller, we need to determine gains K_1 , K_2 , and K_3 .

$$G_{PID}(s) = \frac{K(s+55.92)(s+0.5)}{s} = \frac{4.6(s^2+56.42s+27.96)}{s}$$

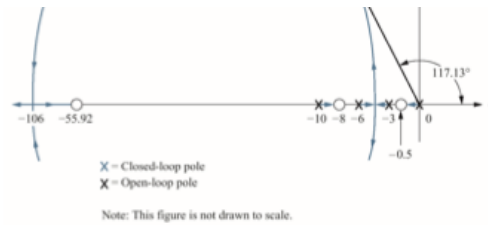
- ▶ Equating with PID controller equation, we have $K_3 = 4.6$.



$$\frac{K_1}{K_3} = 56.42, \text{ and } \frac{K_2}{K_3} = 27.96.$$

- Thus $K_1 = 259.5$ and $K_2 = 128.6$.

Figure 9.36



PID Controller Design Example V

- Simulation compares original, PD compensated, and PID compensated system.
- PID compensated system takes 3 seconds to get to unity.
- If need faster, redesign PD controller.

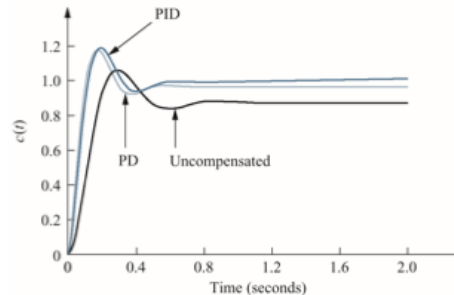


Figure 9.35

Lag-Lead Compensator Design

1. Evaluate performance of original system.
2. Design lead compensator for transient response improvement.
3. Simulate system to be sure requirements met.
4. Redesign if simulation shows they are not.
5. Evaluate steady state error performance of lead compensated system.
6. Design lag compensator for steady state error requirement.
7. Simulate system to be sure all conditions met.

8. Redesign if not met.

SS 10. Consider the unity-feedback system of [Figure P9.1](#) with

$$G(s) = \frac{K}{s(s+20)(s+40)}$$

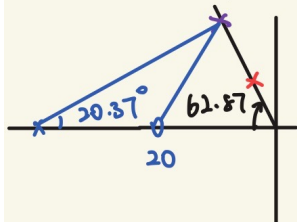
The system is operating at 20% overshoot. Design a compensator to decrease the settling time by a factor of 2 without affecting the percent overshoot and do the following: [[Section: 9.3](#)]

- Evaluate the uncompensated system's dominant poles, gain, and settling time.
- Evaluate the compensated system's dominant poles and settling time.
- Evaluate the compensator's pole and zero. Find the required gain.
- MATLAB ML** Use MATLAB or any other computer program to simulate the compensated and uncompensated systems' step response.

Q10.

$$G(s) = \frac{1}{s^3 + 60s^2 + 800s}, \text{ type 1. Req: 20\% of 0.5\%, } T_s = \frac{T_{s0}}{2}$$

$$(a) \xi = -\frac{\theta_n(0.5\%/100)}{\sqrt{\pi^2 + \ln^2(\frac{0.5\%}{1\%})}} = 0.4559, \theta = \arccos(\xi) = 62.87^\circ$$



so we find dominant pole / working point
 $= -6.39 + 12.5j$ for uncompensated system.
 $K = 9.31 \times 10^3$

$$\therefore T_{s0} = \frac{4}{\sigma_{d0}} = 0.626s \quad s_{d0} = -6.39 + 12.5j$$

$$(b) T_s = \frac{1s}{2} = 0.213 s$$

$\therefore$ we want find new working point $s_d = -\sigma_d + \omega_d j$

$$\therefore \sigma_d = \frac{4}{T_s} = 12.78$$

$$\therefore \frac{\omega_d}{\sigma_d} = \tan 62.87^\circ \therefore \omega_d = \sigma_d \cdot \tan 62.87^\circ = 24.95$$

$$\therefore s_d = -12.78 + 24.95j$$

$$(c) \text{ let } G_c = \frac{s+2}{s+p}$$

$$\angle G(s_d) = 126.51^\circ, \text{ We want } \angle (G_{pd} \cdot G)(s_d) = (2k+1)180$$

$$\text{let } k=0 \therefore \angle G_{pd}(s_d) + \angle G(s_d) = 180 \therefore \angle G_{pd} = 180 - 126.51$$

$$\text{We make } z=20 \therefore \text{now } \angle G_c(s_d) = -159.63 = 53.49^\circ$$

$$\text{so: } \angle G_1(s_d) - \angle G_p = -180$$

$$\therefore \angle G_p = \angle G_1(s_d) + 180^\circ = 20.37^\circ$$

$$\therefore \tan(\theta_{Gp}) = \frac{\omega_d}{p - \sigma_d} \therefore p - \sigma_d = \frac{\omega_d}{\tan \theta_{Gp}}$$

$$\therefore p = \frac{\omega_d}{\tan \theta_{Gp}} + \sigma_d = 80$$

$$\therefore G_c(s) = \frac{s+20}{s+80}$$

$$\text{Define } G_{overall} = G_c \cdot G \text{ and } k = \frac{1}{|G_{overall}(s_d)|}$$

$$k = 7.42 \times 10^4$$

```

s=tf('s');
G=1/(s*(s+20)*(s+40));
disp('G=');
tf(G)
POS=20;
zeta=-log(POS/100)/sqrt(pi^2+(log(POS/100))^2);
disp('zeta=');
disp(zeta);
theta=rad2deg(acos(zeta));
disp('theta=');
disp(theta);
% rlocus(G)
% sgrid(zeta,[])
sigma_do=6.39;
w_do=12.5;
s_do=-sigma_do+1j*w_do;
T_so=4/sigma_do;
disp('T_so=');
disp(T_so);
T_s=T_so/2;
disp('T_s=');
disp(T_s);
sigma_d=4/T_s;
disp('sigma_d=');
disp(sigma_d);
w_d=sigma_d*tand(theta);
disp('w_d=');
disp(w_d);
s_d=-sigma_d+1j*w_d;
angle_G=rad2deg(angle(evalfr(G,s_d)));
disp('angle_G=');
disp(angle_G);
angle_PD=180-angle_G;
disp('angle_PD=');
disp(angle_PD);
G1=G*(s+20);
angle_G1=rad2deg(angle(evalfr(G1,s_d)));
disp('angle_G1=');
disp(angle_G1);
angle_G_P=angle_G1+180;
disp('angle_G_P=');
disp(angle_G_P);

```

```

p=w_d/tand(angle_G_P)+sigma_d;
disp('p=');
disp(p);
z=20;
Gc=(s+z)/(s+p);
G_overall=G*Gc;
k=1/abs(evalfr(G_overall,s_d));
disp('k=');
disp(k);
k_uncompensated=9.31e3;
T_uncompensated=feedback(k_uncompensated*G,1);
T_compensated=feedback(k*G_overall,1);
step(T_uncompensated,T_compensated)
legend('Uncompensated','Compensated')

```

Problem

Given the plant

$$G(s) = \frac{20(s+5)}{s(s+1)(s+4)}$$

design the phase-variable feedback gains to yield 9.5% overshoot and a settling time of 0.74 seconds.

Solution

Step 1

$$\%OS = 9.5\%$$

$$\zeta = \frac{-\ln(\%OS/100)}{\sqrt{\pi^2 + \ln^2(\%OS/100)}}$$

$$\zeta = 0.6$$

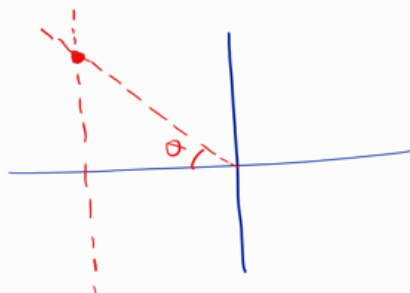
$$\theta = \cos^{-1} \zeta$$

$$\theta = 53.13^\circ$$

$$T_s = 0.74s = \frac{4}{\sigma_d}$$

$$\sigma_d = \frac{4}{0.74} = 5.4$$

$$\tan 53.13^\circ = \frac{\omega_d}{5.4}$$



$$\omega_d = 7.2$$

T_s

$$\tan \theta = \frac{\omega_d}{\sigma_d}$$

$$\therefore \text{desired pole: } s_d = -5.4 \pm 7.2j$$

3rd order system, must have third pole
This should be at $s = -5$ to cancel
the finite zero but let's say it is
slightly off at $s = -5.1$

$\therefore$ the characteristic equation will be

$$(s+5.1)(s+5.4+7.2j)(s+5.4-7.2j) = 0$$

$$s^3 + 15.9s^2 + 136.08s + 413.1 = 0 \quad (1)$$

Step 2

$$G(s) = \frac{20(s+5)}{s(s+1)(s+4)} = \frac{20s+100}{s^3+5s^2+4s}$$

Plant

$$\dot{\underline{x}} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & -4 & -5 \end{bmatrix} \underline{x} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} r$$

$$y = [100 \ 20 \ 0] \underline{x}$$

Closed Loop System

$$r = -\underline{K} \underline{x} \quad \left(\text{closed loop system by feeding back each state variable to input} \right)$$

$$\underline{A} - \underline{B}\underline{K} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -k_1 & -(4+k_2) & -(5+k_3) \end{bmatrix}$$

$$\det(s\underline{I} - (\underline{A} - \underline{B}\underline{K})) = s^3 + (5+k_3)s^2 + (4+k_2)s + k_1 = 0 \quad (2)$$

Equate (1) and (2)

This gives $k_1 = 413.1$

$$k_2 = 132.08$$

$$k_3 = 10.9$$

$\therefore$ we get the following state space representation of the closed loop system

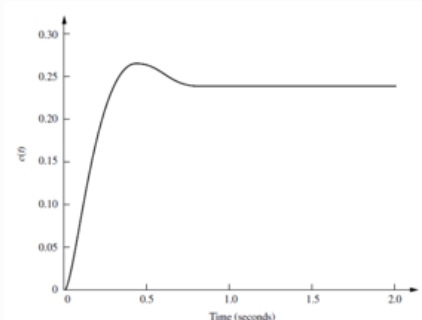
$$\underline{\dot{x}} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -413.1 & -136.08 & -15.9 \end{bmatrix} \underline{x} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} r$$

The transfer function is

$$T(s) = \frac{20(s+5)}{s^3 + 15.9s^2 + 136.08s + 413.1}$$

$\%OS = 11.5\%$
 $T_s = 0.8s$

} - Does not meet our criteria
 - move third pole to $s = -5$ to meet the requirements



Note: simulation shows a steady state value of 0.24 instead of unity
 $\therefore$ very large steady state error, transient properties are not the only important criteria.

第一步：评估原系统并确定期望的主导极点 (s_d)

这一步的目的是根据题目要求的性能指标（超调量 %OS、峰值时间 T_p 或调节时间 T_s ），计算出我们希望系统闭环极点移动到的确切位置。

1. 算阻尼比 ζ ：根据题目给定的期望超调量 %OS，利用公式 $\zeta = \frac{-\ln(\%OS/100)}{\sqrt{\pi^2 + \ln^2(\%OS/100)}}$ 算出阻尼比 2。

2. 算射线角度 θ ：计算复平面上的射线角度 $\theta = \cos^{-1}(\zeta)$ （即射线与负实轴的夹角，或利用 $\tan \theta$ 关系） 2。

3. 算目标频率参数：

◦ 如果题目要求新的峰值时间 T_p ，利用 $T_p = \frac{\pi}{\omega_d}$ 算出期望的阻尼振荡频率 ω_d （即极点的

虚部高度) 2。

。如果题目要求新的调节时间 T_s ，利用 $T_s = \frac{4}{\sigma_d}$ 算出期望的实部绝对值 σ_d 3。

4. 确定期望极点坐标 s_d ：结合算出的 ω_d 和射线角度（例如通过 $\tan \theta = \frac{\omega_d}{\sigma_d}$ ），得出期望的闭环主导极点坐标 $s_d = -\sigma_d \pm j\omega_d$ 2。

第二步：设计 PD 控制器部分 (改善瞬态)

这一步的目的是通过添加一个零点，利用“相角条件”强行让根轨迹穿过我们刚刚算出的 s_d 。

1. 计算原系统相角：将算出的 s_d 代入原开环传递函数 $G(s)$ ，计算所有原系统开环极点和零点到 s_d 的相角总和 $\angle G(s_d)$ 2。
2. 计算需要的补偿角度 θ_z ：设 PD 零点提供的角度为 θ_z ，根据相角条件必须满足 $\angle G(s_d) + \theta_z = (2k + 1)180^\circ$ （通常取 -180° ）。算出 $\theta_z = -180^\circ - \angle G(s_d)$ 2。
3. 定位 PD 零点位置 z_c ：利用几何三角关系 $\tan(\theta_z) = \frac{\omega_d}{z_c - \sigma_d}$ （其中 ω_d 是 s_d 的虚部高度， σ_d 是 s_d 实部的绝对值），解出 PD 零点在实轴上的确切坐标 z_c 2。
4. 写出 PD 补偿器：此时得到 PD 控制器部分为 $G_{PD}(s) = s + z_c$ 2。

第三步：设计 PI 控制器部分 (改善稳态)

这一步的目的是为了消除稳态误差，必须在原点加一个积分器（极点），同时为了不破坏刚才做好的瞬态设计，需紧挨着加一个零点。

1. 加原点极点：增加一个 $1/s$ 的极点，将系统类型（System Type）提升一级（例如从 Type 0 变为 Type 1），从而消除阶跃输入的稳态误差 4，2。
2. 加抵消零点：在极其靠近原点的地方（通常选择 $s = -0.5$ 左右，具体可根据题目习惯）加一个零点，用来抵消新极点对远处主导极点 s_d 造成的角度干扰 2。
3. 写出 PI 补偿器：此时得到 PI 控制器部分为 $G_{PI}(s) = \frac{s+0.5}{s}$ 2。

第四步：计算系统的总综合增益 (K)

此时系统的总等效开环传递函数为 $G_e(s) = K \cdot G_{PI}(s) \cdot G_{PD}(s) \cdot G(s)$ 。我们需要计算出让极点精确落在 s_d 所需的 K 值。

1. 使用幅值条件：根据闭环特征方程 $1 + K \cdot G_{PI}(s_d) \cdot G_{PD}(s_d) \cdot G(s_d) = 0$ ，可知在该点处 $|K \cdot G_e(s_d)| = 1$ 2。
2. 求解 K ：将 s_d 代入除 K 以外的所有部分，算出它们的幅值，然后取倒数即可求出综合增益 $K = \frac{1}{|G_{PI}(s_d) \cdot G_{PD}(s_d) \cdot G(s_d)|}$ 2。

第五步：提取标准的 PID 参数 (K_1, K_2, K_3)

最后，我们需要将算出来的控制器转换回工业标准的 PID 形式，以求出具体的比例、积分、微分增益。

1. 展开补偿器方程：将带入 K 值的总补偿器 $G_c(s) = G_{PI} \cdot G_{PD} \cdot K = \frac{K(s+0.5)(s+z_c)}{s}$ 进行多项式展开，整理成 $\frac{As^2+Bs+C}{s}$ 的形式 2。

2. **对照标准方程：**PID 的标准传递函数形式为 $G_c(s) = K_1 + \frac{K_2}{s} + K_3 s = \frac{K_3 s^2 + K_1 s + K_2}{s}$ （其中也可以提出 K_3 写成 $\frac{K_3(s^2 + \frac{K_1}{K_3}s + \frac{K_2}{K_3})}{s}$ ）¹。

3. **对比系数得出结果：**对比分子多项式的系数，严谨地得出对应关系²：

- K_3 (微分增益 Derivative) = 二次项 s^2 的系数
- K_1 (比例增益 Proportional) = 一次项 s 的系数
- K_2 (积分增益 Integral) = 常数项系数